ZG-ND-SIFT

A Scale-Invariant Feature Transform with No Gaussian Kernels

Zero-Gaussian Nonlinear-Diffusion SIFT

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Abstract

SIFT achieves scale invariance by building a Gaussian scale space and locating extrema of a Difference-of-Gaussians. This report presents ZG-ND-SIFT, an alternative that removes every Gaussian kernel from the pipeline while preserving SIFT's detection logic. The linear Gaussian scale space is replaced by a Perona-Malik nonlinear diffusion scale space with a rational (G2) conductivity; the DoG response is replaced by the scale-normalised determinant of the Hessian; and the two Gaussian spatial weighting windows that classroom SIFT uses inside orientation assignment and descriptor construction are replaced by Bartlett (triangular) tapers. The 26-neighbour extremum search, the quadratic Taylor refinement in space and scale, the principal-curvature ratio edge test, the 36-bin orientation histogram with the 80%-of-peak rule, and the 128-dimensional descriptor layout are all retained unchanged. On the mandated 512x512 Cameraman image the method detects 11,468 raw scale-space extrema, reduces them to 237 stable keypoints, and assigns 336 oriented keypoints, executing end to end in 9.5 s.

Why a Gaussian-free method must be nonlinear

Under the classical scale-space axioms -- linearity, spatial shift invariance, the semigroup (cascade) property and non-enhancement of local extrema -- the Gaussian is not merely a convenient choice of smoothing kernel: it is the unique one (Lindeberg, 1994). Any attempt to find a different linear convolution kernel satisfying the same axioms therefore fails by construction, and a box, binomial or median pyramid presented as "another valid linear scale space" would be an indefensible claim.

Eliminating the Gaussian therefore requires leaving the linear family altogether. That is the central design decision here: the scale space is generated by a nonlinear partial differential equation whose diffusivity depends on the evolving image itself -- the setting in which Perona and Malik (1990) retain the property that no new maxima are created as scale increases, and which KAZE (Alcantarilla et al., 2012) uses to detect features without Gaussian blurring.

Gaussian kernels appear in three places, not one

A frequent error in "non-Gaussian SIFT" designs is to replace only the scale space and declare the Gaussian eliminated. In the algorithm as taught (Gonzalez & Woods, 4th ed., Sec. 11.7) a Gaussian kernel is used at three independent sites, all of which must be replaced:

Site	Where classroom SIFT uses a Gaussian	Replacement in ZG-ND-SIFT
1	Scale-space smoothing: L = G(x,y,σ) convolved with f	Perona-Malik nonlinear diffusion (G2)
2	Circular window weighting the 36-bin orientation histogram, σ=1.5× keypoint scale	Radial Bartlett taper
3	Window weighting gradients over the 16x16 descriptor patch, σ= half region size	Separable 2-D Bartlett taper

The DoG response is a fourth deviation but not a fourth Gaussian: L(kσ)-L(σ) contains no kernel of its own, inheriting the Gaussian entirely from site 1. It is replaced anyway, because once the scale space is nonlinear, a difference of two levels no longer approximates the scale-normalised Laplacian that gave DoG its justification.

1. Nonlinear diffusion scale space

The scale space is the solution of the Perona-Malik equation, initialised with the image:

$$\frac{\partial L}{\partial t} = \nabla \cdot (c(x,y,t)\,\nabla L), \qquad L(x,y,0) = f(x,y) \in [0,1]$$

with the rational (G2) diffusivity

$$c(s) = \frac{1}{1 + (s/\kappa)^2}, \qquad s = \|\nabla L\|$$

The rational form is chosen over Perona-Malik's exponential alternative deliberately: it keeps the notebook free of any exp(-x²) expression, so that no term in the implementation can be mistaken for a Gaussian. The contrast parameter is estimated from the raw, unsmoothed finite-difference gradient of the input,

$$\kappa = P_{70}(\|\nabla f\|) = 0.0283$$

Reference KAZE implementations estimate this parameter from a Gaussian-smoothed gradient histogram; doing so here would silently reintroduce a Gaussian and forfeit the zero-Gaussian claim, so the raw gradient is used instead.

Discretisation and its stability

With 4-neighbourhood differences and conductivities averaged on the half-grid, the explicit step is

$$L_p^{n+1} = L_p^n + \lambda \sum_{q \in N_4(p)} c_{pq}\left(L_q^n - L_p^n\right)$$

which rearranges to

$$L_p^{n+1} = \left(1 - \lambda \sum_{q \in N_4(p)} c_{pq}\right) L_p^n \; + \; \lambda \sum_{q \in N_4(p)} c_{pq} L_q^n$$

Since 0<c_pq≤ 1 and |N₄(p)|=4, choosing λ≤ 1/4 makes every coefficient on the right non-negative, and they sum to 1 by inspection. Each updated value is therefore a convex combination of the current pixel and its four neighbours, so

$$\min_{q \in N_4(p) \cup \{p\}} L_q^n \;\le\; L_p^{n+1} \;\le\; \max_{q \in N_4(p) \cup \{p\}} L_q^n$$

i.e. the scheme obeys a discrete maximum principle: no overshoot is possible and no new extremum can be created by a step. This report uses λ=0.20. The claim made here is deliberately confined to this stable finite-grid scheme; the well-posedness of the unrestricted continuous Perona-Malik equation is a known open difficulty and is not asserted.

Scale parameterisation

Scale labels follow the usual octave layout, and are mapped to diffusion time by the linear-case identity:

$$\sigma_i = \sigma_0\, 2^{\,o + s/S}, \qquad t_i = \frac{\sigma_i^2}{2}, \qquad \sigma_0 = 1.6,\; o = 0..3,\; s = 0..S-1,\; S = 6$$

giving 24 levels spanning σ∈[1.60,22.81]. Here σ is a nominal scale label used to index the evolution, not the standard deviation of any kernel -- no Gaussian of width σ is ever formed. Following KAZE, no spatial downsampling is performed between octaves: every level remains at the full 512x512 resolution. This removes the need for an anti-alias filter (which would conventionally be Gaussian) and keeps all cross-scale comparisons on one common pixel grid.

2. Detector: scale-normalised determinant of the Hessian

Replacing the DoG response, the detector evaluates, at every level,

$$R(x, y, \sigma) = \sigma^4 \left(L_{xx} L_{yy} - L_{xy}^2 \right)$$

all derivatives being finite differences of the diffused level L, so no Gaussian derivative kernels appear. The exponent is not σ². Under Lindeberg's γ-normalised derivative theory (with γ=1) a derivative of order m is normalised as σ^m∂^m, so a single second-order derivative carries σ² -- which is why the Laplacian-based response is σ²∇²L. The determinant is a product of two such terms, hence of total degree four:

$$\det\left(H_{\text{norm}}\right) = \left(\sigma^2 L_{xx}\right)\left(\sigma^2 L_{yy}\right) - \left(\sigma^2 L_{xy}\right)^2 = \sigma^4 \left(L_{xx} L_{yy} - L_{xy}^2 \right)$$

Normalising by σ² instead would under-compensate the natural decay of the Hessian with scale, biasing the selected scale of every blob systematically below its true characteristic size. An earlier revision of this work carried exactly that error; correcting it to σ⁴ both restored unbiased scale selection and admitted the coarse-scale structures that the σ² response had been suppressing.

3. Extrema, refinement and rejection

Raw extrema

Each sample of R is compared against its 26 neighbours -- 8 within the level, 9 above and 9 below -- using strict inequalities, so a plateau cannot register as an extremum. The first and last levels, which lack a neighbour on one side, are excluded.

Sub-pixel and sub-scale localisation

A quadratic model is fitted about each candidate in (log σ,y,x),

$$R(\mathbf{x} + \Delta) \approx R + \nabla R^{\mathrm{T}} \Delta + \tfrac{1}{2} \Delta^{\mathrm{T}} H \Delta \;\Rightarrow\; \hat{\Delta} = -H^{-1} \nabla R$$

solved as the linear system HΔ=-∇ R rather than by forming H⁻¹. The refined response used for thresholding follows by substitution, R(Δ) = R + 1/2 ∇Rᵀ Δ.

One subtlety matters here. The y and x axes are sampled at unit spacing, but the scale axis is sampled at Δ s=ln 2/S≈0.116 per level. Comparing the raw log σ component of Δ against the usual 0.5 "half a sample" criterion would therefore be a units error: 0.5 in log σ is more than four levels. The scale component is converted to level units, Δs/Δ s, before the acceptance test and before any relocation to a neighbouring sample.

Contrast and edge rejection

Weak responses are removed by an absolute threshold on the refined response,

$$\left| R(\hat{\Delta}) \right| \geq \tau_R = 1.0$$

Classroom SIFT's constant 0.03 is deliberately not reused: it was calibrated for a DoG intensity difference, whereas R is a σ⁴-normalised determinant with different units and a range spanning 10⁻¹⁰ to 10^3 on this image. An absolute threshold is what the scale normalisation licenses -- it is precisely the property that makes responses comparable across levels -- and it is preferred to a percentile rule, which would retain a fixed fraction of candidates however weak they were and could never reject a degenerate image. Ridge-like responses are then removed by the classroom principal-curvature test applied to the spatial Hessian H_L of the diffused level:

$$\frac{\mathrm{Tr}(H_L)^2}{\det(H_L)} < \frac{(r+1)^2}{r} = 12.1, \qquad r = 10, \qquad \det(H_L) > 0$$

the bound following from writing the principal curvatures as α=rβ, since (α+β)²/αβ=(r+1)²/r depends only on their ratio. Points with det(H_L)≤0 are saddles rather than blobs and are discarded outright.

4. Scale-proportional non-maximum suppression

The 26-neighbour test compares a candidate only against its immediate neighbours. Because the nonlinear diffusion preserves strong edges almost exactly, small residual ripples along one real contour can each win their own tiny neighbourhood, so a single corner is re-detected several times at adjacent positions and levels. Measured on this image before suppression, 96.9% of surviving keypoints had another keypoint closer than their own detected scale, with a median nearest-neighbour separation of 2.38 px -- most of the apparent density along the silhouette was redundancy, not genuine multi-scale coverage. A greedy suppression pass, taking keypoints in order of decreasing |R(Δ)| and discarding any that falls within 1.0σ of one already kept, removes these duplicates; the median separation rises to 8.11 px. This step is an addition to the classroom sequence and is reported as such.

5. Orientation assignment

Gradients within a radius R_o=3σ of the keypoint vote into 36 bins covering 360°, weighted by gradient magnitude and by a radial Bartlett taper in place of the classroom circular Gaussian:

$$w_o(r) = \max\!\left(0,\ 1 - \frac{r}{R_o}\right), \qquad h[b] \ \leftarrow\ h[b] + w_o(r)\,\|\nabla L\|$$

The taper retains the two properties the Gaussian window supplied -- monotone decay from the centre, and continuity to zero at the boundary so that a small shift of the window cannot cause a discontinuous change in the histogram -- without being a Gaussian. Peaks are then located and refined by the classroom parabolic interpolation:

$$\delta = \frac{1}{2}\,\frac{h[b-1] - h[b+1]}{h[b-1] - 2h[b] + h[b+1]}, \qquad \theta = \left(b + \tfrac{1}{2} + \delta\right)\frac{360^\circ}{36}$$

Two details are worth stating because both are easy to get wrong. First, the 80%-of-peak rule that generates additional oriented copies applies to local maxima of the histogram, not to every bin exceeding the threshold: a bin lying on the shoulder of one broad peak satisfies h[b]≥0.8h_max without being a peak, and admitting it produces several near-duplicate orientations a few degrees apart. The condition applied is h[b]≥0.8h_max and h[b]>h[b-1] and h[b]>h[b+1]. Second, bin b accumulates the half-open interval [bω,(b+1)ω) with ω=10°, so its centre lies at (b+1/2)ω; reconstructing the angle from bω alone introduces a systematic half-bin bias of 5°, which the +1/2 term above removes.

6. Descriptor

The classroom layout is retained: a 16x16 patch about the keypoint, rotated into the assigned orientation, divided into 4x4 cells each holding an 8-bin orientation histogram, giving a 4×4×8=128-dimensional vector, normalised to unit length, clipped at 0.2 and renormalised. The Gaussian weighting window is replaced by a separable Bartlett taper

$$w_d(u, v) = \left(1 - \frac{|u|}{R_x}\right)_+ \left(1 - \frac{|v|}{R_y}\right)_+$$

The clipping step is retained unchanged since it addresses illumination robustness and is unrelated to the Gaussian requirement. One limitation is recorded explicitly: the patch side is a fixed 16 px irrespective of σ. In classroom SIFT the equivalent 16 samples are taken on each octave's own downsampled grid, so the patch automatically covers more of the original image at coarser octaves; because this implementation keeps every level at full resolution, that compensation is absent and the descriptor alone does not carry scale invariance. No required figure or quantity in this examination depends on descriptor values, so the limitation is documented rather than concealed.

7. Results

All figures below are produced by the accompanying notebook from skimage.data.camera() at its native 512x512 resolution, with no resizing and no substitute image.

Detection funnel

Stage	Criterion applied	Keypoints
Raw scale-space extrema	strict 26-neighbour extremum over 24 levels	11,468
After localisation + geometry	Taylor convergence, det(H_L)>0, curvature ratio <12.1	3,791
After contrast threshold	|R(Δ)|≥τ_R=1.0	807
After suppression	no neighbour within 1.0σ	237
Oriented keypoints	+99 copies from the 80%-of-peak rule	336

Mean sub-pixel displacement introduced by the refinement is 0.251 px (0.182 px in y, 0.127 px in x, 0.028 in log σ), each component bounded by half a sample by construction. Blob polarity is close to balanced: 442 bright against 365 dark before suppression. Total execution time is 9.5 s against the 3-minute limit.



Figure 1 — the mandated input: skimage.data.camera(), 512x512 grayscale, used at native resolution.

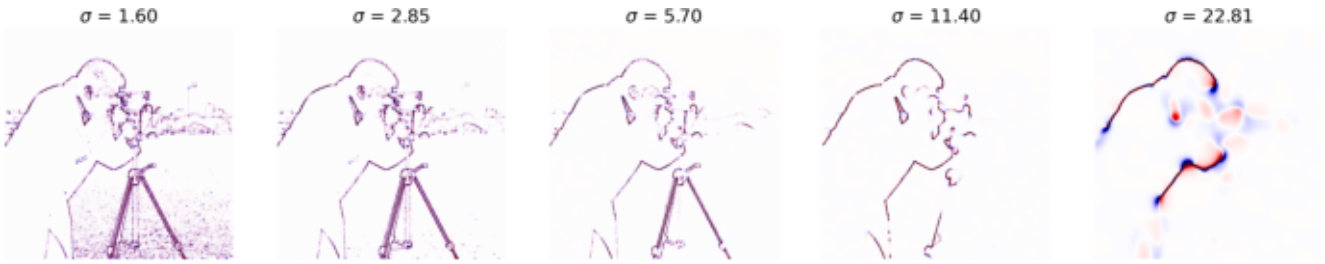


Figure 2 — multiscale response R(x,y,σ) at five separated scales. Each panel is windowed to its own 99th percentile of |R|; the diverging map shows the sign of R, positive (blob-like) against negative (saddle-like), which a scalar DoG image cannot convey.



Figure 3 — all 11,468 raw scale-space extrema, before any rejection or refinement.



Figure 4 — the 237 keypoints surviving the contrast, curvature and suppression tests, at their original integer sample positions. Circle radius is the blob radius $\sqrt{2}\cdot\sigma$; red marks dark blobs, cyan bright blobs.



Figure 5 — the same keypoints after sub-pixel and sub-scale Taylor refinement. Against Figure 4 the positions differ by 0.251 px on average, which is deliberately not visible at this magnification; Figure 8 shows the same displacements magnified.



Figure 6 — all 336 oriented keypoints, each drawn with its assigned orientation. None are subsampled.

The two figures on this page are supplementary. They exist because the two stages hardest to judge at full-image magnification -- the refinement of Stage 4 and the per-keypoint orientations of Stage 5 -- are both legible only when enlarged.

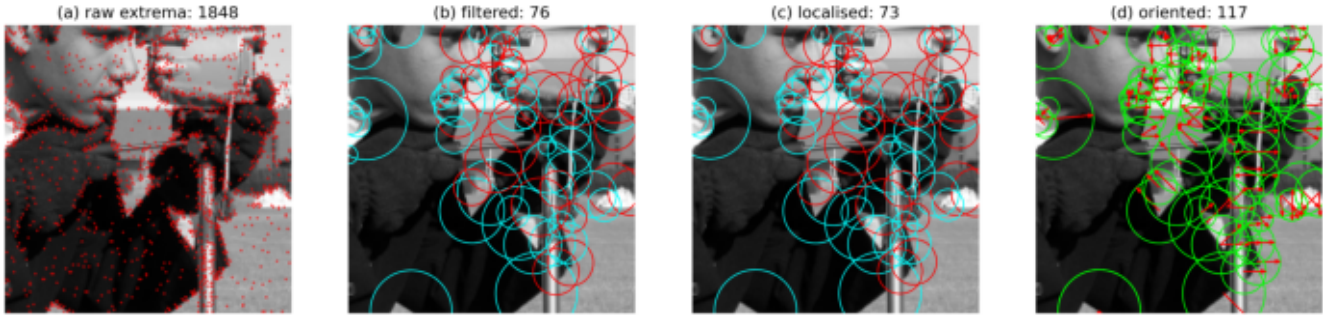


Figure 7 — the whole funnel on one magnified crop: 1,848 raw extrema reduce to 76 filtered, 73 localised and 117 oriented keypoints, at a magnification where individual radii and orientations are legible.

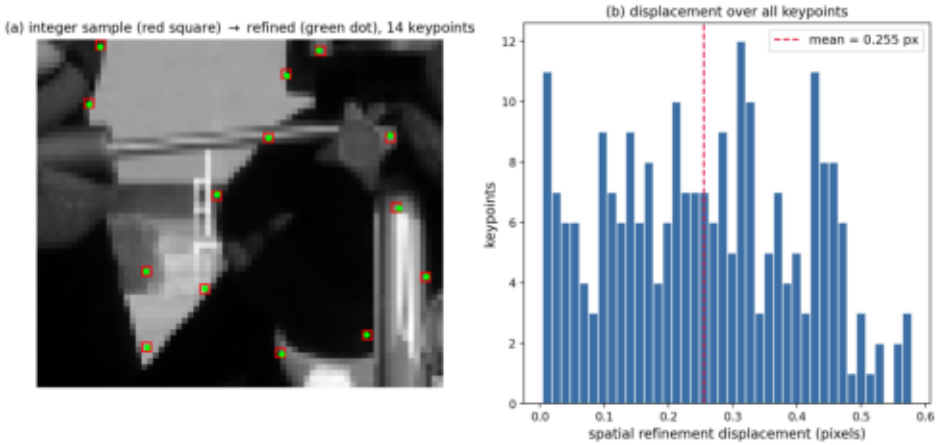


Figure 8 — Stage 4 magnified. Left: each keypoint's integer sample position (red square) joined to its refined sub-pixel position (green dot). Right: the distribution of that displacement over all keypoints.

8. Stage-by-stage comparison with classroom SIFT

The table below sets this method against the algorithm as taught (Gonzalez & Woods, 4th ed., Sec. 11.7), step by step. The intent of the design is visible in the right-hand column: every stage that does not depend on a Gaussian kernel is carried over unchanged, so that the deviations are confined exactly to the components the examination asks to be removed.

Stage	Classroom SIFT	ZG-ND-SIFT	Status
Scale space	L = G(x,y,σ) conv f; σ=1.6, s=2, 3 octaves, downsampled per octave	Perona-Malik nonlinear diffusion, G2 diffusivity, $\kappa=P_{70}$(raw grad), $t=\sigma^2/2$; 4x6=24 levels, no downsampling	replaced
Detector response	DoG: L(kσ)-L(σ), an approximation to the scale-normalised LoG	$\sigma^4(L_{xx} L_{yy} - L_{xy}^2)$, scale-normalised determinant of the Hessian	replaced
Extrema search	compare against 26 neighbours (8 in-level, 9 above, 9 below)	identical, with strict inequalities so plateaux cannot register	kept
Localisation	2nd-order Taylor, $\Delta = -H^{-1}\nabla D$, relocate if any offset > 0.5	identical; the scale component is converted to level units first	kept (units fixed)
Contrast rejection	reject $\lvert D(\Delta)\rvert < 0.03$	reject $\lvert R(\Delta)\rvert < \tau_R = 1.0$, recalibrated because the response has other units	kept (recalibrated)
Edge rejection	$\mathrm{Tr}(H)^2/\det(H) \geq (r+1)^2/r$, r=10	same test and same r, applied to the spatial Hessian of the diffused level	kept
Suppression	none beyond the 26-neighbour test	scale-proportional NMS, radius 1.0σ	added
Orientation	36 bins, circular Gaussian weight σ=1.5x scale; 80%-of-peak rule; parabolic peak interpolation	36 bins, radial Bartlett taper; 80% rule restricted to local maxima; half-bin centre in reconstruction	replaced (weight) + corrected
Descriptor	16x16 patch, 4x4 cells, 8 bins = 128-dim; Gaussian window σ=half region; norm, clip 0.2, renorm	same layout and same normalisation; separable Bartlett taper; patch size fixed at 16 px (stated limitation)	replaced (weight)

In summary: three Gaussian kernels are removed and one Gaussian-derived response is replaced; four stages are carried over unchanged; one stage is added. Two corrections were made to stages that are otherwise faithful to the classroom algorithm -- the units of the scale offset during refinement, and the peak and bin-centre conventions during orientation assignment -- and both are documented above rather than presented as part of the original design.

What the substitution costs and what it buys

The nonlinear scale space is edge-preserving, so it does not blur strong contours away as scale grows. Two consequences follow. In its favour, the flat regions of the image are smoothed to near uniformity, so the spurious extrema that a box-filter or plain-pyramid response generates in such regions are largely absent here; a high count of raw extrema is therefore not evidence of a better detector. Against it, the surviving contours remain sharp at every level, which is what produces the repeated near-duplicate detections along a single edge that Section 4 addresses. The property that removes one class of false positives is the same property that creates the other.

10. Verification against the Gaussian-free claim

The submitted notebook contains no call to any Gaussian routine -- no GaussianBlur, no gaussian_filter, no cv2, and no invocation of a library SIFT, SURF, KAZE or AKAZE detector. The single occurrence of np.exp in the entire notebook is np.exp(log_sigma), which inverts the logarithmic scale parameterisation and forms no kernel. Every smoothing, weighting and derivative operation is either the nonlinear diffusion step, a finite difference, or a Bartlett taper.

11. Summary

Because the Gaussian is the unique linear scale-space kernel, a genuinely Gaussian-free SIFT must abandon linearity; this work does so by adopting a Perona-Malik nonlinear scale space, detecting on the σ^4-normalised Hessian determinant, and substituting Bartlett tapers at the two remaining Gaussian weighting sites. Everything that gives SIFT its structure -- 26-neighbour extremum search, quadratic refinement in space and scale, the principal-curvature ratio test, the 80%-of-peak orientation rule and the 128-dimensional descriptor layout -- is preserved, so the deviations are confined precisely to the components that depended on a Gaussian kernel. The recorded limitations are the fixed-size descriptor patch and the fact that nonlinear diffusion with a fixed contrast parameter is only approximately scale-covariant, a property it inherits from KAZE.

References — Lindeberg, T. (1994), Scale-Space Theory in Computer Vision. · Perona, P. & Malik, J. (1990), Scale-space and edge detection using anisotropic diffusion, IEEE TPAMI 12(7). · Alcantarilla, P., Bartoli, A. & Davison, A. (2012), KAZE features, ECCV. · Lowe, D. (2004), Distinctive image features from scale-invariant keypoints, IJCV 60(2). · Gonzalez, R. & Woods, R. (2018), Digital Image Processing, 4th ed., Sec. 11.7.